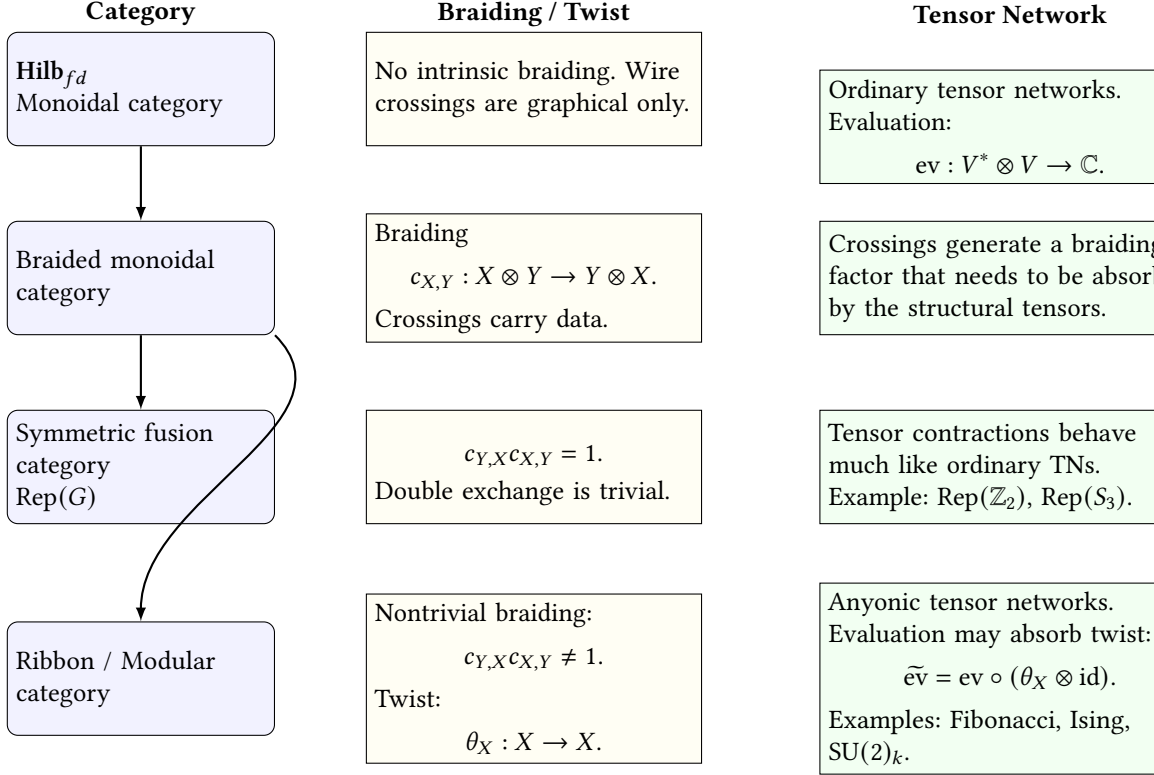


1 DECOMPOSITIONS FOR CATEGORICAL NETWORKS

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group.



2 DECOMPOSITION OF SEMISIMPLE CATEGORIES

LEMMA 2.1. For a vector space W , there is an isomorphism

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

PROOF. Let $e_{ii \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \rightarrow \bigoplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \dots, w, \dots, 0)$$

and linear, thus

$$\Phi \left(\sum_i c_i e_i \otimes w_i \right) = (c_1 w_1, \dots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. $\square$

LEMMA 2.2. *Let D be a finite dimensional k -vector space and let V be an object of a k -linear category C . Then*

$$D \otimes V \cong V^{\oplus \dim D}$$

PROOF. Let $n = \dim D$. Take a basis $\{e_i\}$ for D . This yields an isomorphism

$$D \cong \bigoplus_{i=1}^n k \cdot e_i \cong k^n$$

Since in a k -linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_{i=1}^n (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

□

LEMMA 2.3 (ISOTOPIC DECOMPOSITION). *Let C be a semisimple k -linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition*

$$V \cong \bigoplus_a D_a \otimes R_a. \quad (1)$$

where we refer to $D_a := \text{Hom}_C(R_a, V)$ as the degeneracy of the space.

PROOF. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} \bigoplus_{i=1}^{D_a^V} a \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\text{Hom}_C(R_b, -)$ yields

$$\text{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actually need this: Maybe this is simpler if we always let the degeneracy be C ?

LEMMA 2.4. *Let C be a k -linear category and let D, E be finite-dimensional k -vector spaces. For any objects $R, S \in C$, there is a natural isomorphism*

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S).$$

PROOF. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_C(R^{\oplus m}, S^{\oplus n}) \quad \text{by lemma 2.1}$$

$$\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \text{Hom}_C(R, S) \quad \text{by additivity of } \text{Hom}_C$$

$$\cong k^{mn} \otimes \text{Hom}_C(R, S) \quad \text{by lemma 2.1}$$

$$\cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S) \quad \text{since } \text{Hom}_k(D, E) \cong k^{mn}$$

(The last as s)

□

Definition 2.5 (Copower). For a Vect_k -enriched (and tensored?) category C , and $E \in \text{Vect}_k, X \in C$ the power X^E is the object of C representing the natural isomorphism

$$\text{Hom}_C(Y, X^E) \cong \text{Hom}_k(E, \text{Hom}_C(Y, X))$$

THEOREM 2.6 (DECOMPOSITION OF SEMISIMPLE CATEGORIES). *Let C be a semisimple, monoidal, k -linear category whose tensor product is k -linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of objects in C . Every morphism*

$$T \in \text{Hom}_C \left(\bigotimes_{i=1}^n V_i, \bigotimes_{i=1}^m W_i \right)$$

admit a decomposition as

$$T \in \bigoplus_{\hat{a}, \hat{b}} P^{\hat{a}, \hat{b}} \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \dots, a_n)$ and P is a k -vector space of dimension ???

PROOF.

$$\begin{aligned} & \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_j W_j \right) \\ & \cong \text{Hom}_C \left(\bigotimes_i \left(\bigoplus_{a_i} D_{i, a_i} \otimes R_{a_i} \right), \bigotimes_j \left(\bigoplus_{b_j} E_{j, b_j} \otimes R_{b_j} \right) \right) \quad \text{by lemma 2.3} \\ & \cong \text{Hom}_C \left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}} \right) \quad \text{since } \otimes \text{ is } k\text{-linear in each variable} \\ & \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) \quad \text{by additivity of } \text{Hom}_C \\ & \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}) \quad \text{by lemma 2.4.} \end{aligned}$$

□

COROLLARY 2.7 (OPERATORS ARE BLOCK MATRICES). *This is also given here: [KZ22, Remark 3.4.8]*
An operator

$$\text{Hom}(V, W) \cong \text{Hom}_C \left(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b} \right) \cong \bigoplus_{a, b} \text{Hom}(D_a, E_b) \otimes \text{Hom}_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Hom}(D_a, E_a) \cong \bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Mat}_{d_a \times e_a}(k)$$

COROLLARY 2.8. *Let $Q_{\mu, \eta}$ be a basis of $\text{Hom}_C(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}})$ where μ denotes the multiplicities and η the internal fusions. In a multiplicity-free Category we will omit μ as a label.*

Example 2.9. If we let our monoidal semisimple category just be $\text{Vec}_{\mathbb{C}}$, the only simple object is $\mathbb{C}$. Thus there is also only the trivial biproduct over one sector.

Example 2.10. If we restrict to the category $\text{Rep}(SU(2))$ we can build the symmetric tensor networks as given in [SV12]. TODO: actual ly do this and convince yourself that the notations amd the triplet basis workd as desired.

Example 2.11. Take a 2 in 1 out tensor in $\text{Rep}(SU(2))$ as

$$T \in \text{Hom}_{\text{Rep}(SU(2))}(V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}, V_{\frac{1}{2}})$$

Since all parts are simple, nothing happens.

Example 2.12. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned} & \text{Hom}_C\left(\bigoplus_{a,b,c \in (0,1)} (C \otimes C \otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b \in (0,1)} C \otimes V_b\right) \\ &= \bigoplus_{a,b,c,d \in (0,1)} \text{Hom}_C((C \otimes C \otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\ &= \bigoplus_{a,b,c,d} \text{Hom}(C^3, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d) \end{aligned}$$

Example 2.13. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned} & \text{Hom}_C\left(\bigoplus_{a,b \in (0,1)} (C \otimes C) \otimes V_a \otimes V_b, \bigoplus_{b \in (0,1)} C \otimes V_b\right) \\ &= \bigoplus_{a,b,c \in (0,1)} \text{Hom}_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\ &= \bigoplus_{a,b,c} \text{Hom}(C^2, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b, V_c) \end{aligned}$$

Example 2.14. Consider the same operator as in example 5 but with its incoming legs fused together. So we have $V_{in} = V_{\frac{1}{2}}^4$ and $V_{out} = V_{\frac{1}{2}}^2$. This decomposes $V_2 \oplus V_1^{\oplus 3} \oplus V_0^{\oplus 2}$. Then we can directly use corollary 2.7 to obtain

$$\text{Hom}(V_{in}, V_{out}) =$$

Example 2.15. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ .

$$\begin{aligned} & \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = \text{Hom}_C() \\ & \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) \\ & \cong \text{Hom}_C((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau), k \otimes \tau) \\ & \cong \text{Hom}_C(k \otimes 1 \otimes k \otimes \tau, k \otimes \tau) \oplus \text{Hom}_C(k \otimes \tau \otimes k \otimes \tau, k \otimes \tau) \\ & \cong \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(1 \otimes \tau, \tau) \oplus \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(\tau \otimes \tau, \tau). \end{aligned}$$

Example 2.16 (Anyonic matrix operator). Finally, consider a $4 \rightarrow 4$ anyon operator with one leg on each side, representing the space $\text{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. The full decomposition is $2 + 3\tau$. Hence we obtain

$$\begin{aligned} & \text{Hom}_C((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)) \\ &= \bigoplus_{a,b \in (1,\tau)} \text{Hom}_C(k^{m_a} \otimes a, k^{m_b} \otimes b) = \bigoplus_{a,b \in (1,\tau)} \text{Hom}(k^{m_a}, k^{m_b}) \otimes \text{Hom}_C(a, b) \end{aligned}$$

where

$$m_1 = 2, \quad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= \text{Hom}(k^2, k^2) \oplus \text{Hom}(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

Example 2.17 ($U(1)$ Symmetry). Technically we would want to show here that this is simpler in general. TOOD. Make a theorem? Or a corollary?

Example 2.18 (Ground state on 4 anyons). We can be given a ground state in the operator form as $\text{Hom}(\tau^4, \tau)$.

2.1 Q intertwiners

Lets look at 2 paths for the 4 index tensor, $(a, b, c \rightarrow d)$ and $(a, b \rightarrow c, d)$.

$$(Q_{j_a j_b j_c}^{j_e})_{m_1, m_2, m_3, m_4} = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e; j_3 m_3 | j_4 m_4 \rangle$$

while the other topology would look like

$$Q \dots = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e | j_3 m_3; j_4 m_4 \rangle$$

and as an operator this would have to live in $\text{Hom}(V_{j_1} \otimes V_{j_2}, V_{j_3} \otimes V_{j_4})$

The yoga tree $(-1, 1, -3), (-2, 1, -4)$ Gives us a good mixture of those. This Q would then have to look like

$$Q_{-1-2-3-4}^1 = \sum_1 \langle j_{-1} m_{-1}; j_1 m_1 | j_{-3} m_{-3} \rangle \langle j_{-2} m_{-2} | j_1 m_1; j_{-4} m_{-4} \rangle = \sum_{m_1} C_{-1,1 \rightarrow -3} \cdot C_{-2,1 \rightarrow -4}$$

How would we do this when we have a contraction of two simultaneous indices:

$$Q_{-1-2}^{1,2} = \sum_{1,2} \langle -1 | 1; 2 \rangle \langle 1; 2 | -2 \rangle$$

If we define τ to be the fusion trees topology, then we can define ${}_\tau Q_j^i$

What would this look like for general symmetry groups G ? Lets choose a basis of its invariant decomp $|\lambda_a t_a \mu_a\rangle$. Then the generalized CG coefficient would look like

$$(C_{\lambda_a \lambda_b \rightarrow \lambda_c}^\alpha)_{\mu_a \mu_b}^{\mu_c} = \langle \lambda_a \mu_a; \lambda_b \mu_b | \lambda_c \mu_c \rangle_\alpha$$

where alpha denotes the singular fusion multiplicity at μ_c .

Now to fully specify Q here for k sectors (λ_i, μ_i) , we need to give it $k - 3$ internal edges $(\lambda_{e_1}, \dots, \lambda_{e_{k-3}})$ along with $k - 2$ multiplicity symbols, $(\alpha_1, \dots, \alpha_{k-2})$

Can we write down a general formula for Q ?

This would then be

$$Q_\lambda$$

3 CATEGORIES

3.1 Tensor and Fusion Categories

Let C be a category with bifunctor

$$\otimes : C \times C \rightarrow C, \quad (X, Y) \rightarrow X \otimes Y.$$

and the isomorphism $F : (- \otimes -) \otimes - \simeq - \otimes (- \otimes -)$

... introduce fusion categories here, with associativity and commutativity restrictions, I think this would be a good start.

Definition 3.1 (Monoidal Category). A monoidal category is a category C with a bifunctor $\otimes : C \times C \rightarrow C$

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \rightarrow C \boxtimes D$$

with morphisms given by

$$\text{Hom}_{C \boxtimes D}(X \boxtimes Y, W \boxtimes V) = \text{Hom}_C(X, W) \otimes \text{Hom}_D(Y, V)$$

Definition 3.2 (The Deligne Tensor Product of Categories). Let k be a field, and let C and $\mathcal{D}$ be two k -linear, finite abelian categories. The **Deligne tensor product**, denoted by $C \boxtimes \mathcal{D}$, is a k -linear finite abelian category equipped with a k -bilinear functor

$$\boxtimes : C \times \mathcal{D} \longrightarrow C \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 3.3 (The Category Vec_k). Let k be an algebraically closed field.

- **Objects:** The objects are finite dimensional vectors spaces over k .
- **Morphisms:** Linear maps between vector spaces.

Definition 3.4 (The Category $\text{Rep}_k(G)$). Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\text{Rep}_k(G)$ are pairs (V, ρ_V) , where V is a k -vector space and $\rho_V : G \rightarrow \text{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\text{Rep}_k(G)$, a morphism from V to W is a k -linear transformation $f : V \rightarrow W$ that is G -invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho_V(g)} & V \\ f \downarrow & & \downarrow f \\ W & \xrightarrow{\rho_W(g)} & W \end{array}$$

In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

The set of all such morphisms is denoted by $\text{Hom}_G(V, W)$.

We can further endow $\text{Rep}_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\text{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 3.5 (Direct product of representations). A representation of $G_1 \times G_2$ is a space where the pair (g_1, g_2) acts linearly. SO this is a pair

$$(V, \rho), \rho : (G_1 \times G_2) \rightarrow GL(V)$$

Now, let $(\text{Rep}_{\mathbb{C}}(G), \otimes, \mathbb{C})$ be the monoidal category of representations. We equip the category with a tensor product that acts as follows:

TODO...

THEOREM 3.6 (FUSION REPRESENTATION CATEGORY OF FINITE GROUPS). *Let G be a finite group. Then $\text{Rep}(G)$ is a fusion category.*

THEOREM 3.7. *Let G_1 and G_2 be finite groups and let $\text{Rep}(G_1), \text{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories*

$$\text{Rep}(G_1 \times G_2) \simeq \text{Rep}(G_1) \boxtimes \text{Rep}(G_2)$$

Big issue here: $\text{Rep}(\text{SU}(2))$ is not a fusion category as it is not finite. But if i read it correctly, the deligne product only requires abelianness of its groups. Now, $\text{Rep}(G)$ is monoidal for any G but is it abelian?

So this is actually true, it is abelian. We could obtain a restricted versions by for example always restricting $\text{SU}(2)$ to some $\text{SU}(2)_k$ for some finite k . But i do not yet know how to present this nicely.

A second advantage would be that we can always represent fusion trees as monoidal categories (is this true or is fusion required here again?), and we can write down a unified theory for the fusion trees. Then I hope to write a nice method to view the "eval" of such a fusion tree, i.e. its clebsch gordan tensors by contracting down the topology of the tree.

4 CATEGORIES NEEDED

The motivation for this section is to introduce the necessary category theory background, and provide all the necessary theorems for a format that includes both representations and fusion categories.

Definition 4.1 (Semisimple Category). An abelian category C is semisimple if every object is either simple or a direct sum of simple objects. An object is *simple* if its only subobjects are 0 or X itself. Any object can be written as

$$X = \bigoplus_{i \in I} S_i^{\oplus n_i}$$

LEMMA 4.2 (SCHUR'S LEMMA [Sch05]). *If S and T are simple objects in a semisimple category over an algebraically closed field (this is always C in our case), then it holds that*

$$\text{Hom}(S, T) = \begin{cases} k, & \text{if } S \cong T; \\ 0, & \text{if } S \not\cong T. \end{cases}$$

Proof [IR19, Theorem 7.5].

Lets give a coordinatized approach to fusion categories:

Definition 4.3 (Fusion Category). A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

LEMMA 4.4. Let C be a k -linear monoidal category whose tensor product is k -linear in each variable. Then

$$\bigotimes_{r=1}^n \left(\bigoplus_{a_r} X_{r,a_r} \right) \cong \bigoplus_{\hat{a}} \bigotimes_{r=1}^n X_{r,a_r},$$

where

$$\hat{a} = (a_1, \dots, a_n).$$

LEMMA 4.5 (FINITE GROUPS ADMIT FUSION CATEGORY). Let G be a finite group, the $\text{Rep}(G)$ is a fusion category.

Let $\text{Vec} := (\text{FDCVec}, \otimes, \mathbb{C})$ be the monoidal category of finite dimensional complex vector spaces where the last tuple entry $\mathbb{C}$ denotes the unit object, since there are isomorphisms $V \otimes \mathbb{C} \cong V \cong \mathbb{C} \otimes V$.

Let $(\mathbb{C}\text{Rep}(G), \otimes, ?)$ denote the representation Category for a compact group G . Here $\otimes$ is the tensor product of representations $(\pi \otimes \rho, V \otimes W)$.

Definition 4.6 (Direct sum). Given a category C with objects $\{a_n\}$, an object $\bigoplus_{i=1}^n a_i \in C$ is called the *direct sum*. This is equipped with the injections and projections

$$\iota_i : a_i \rightarrow \bigoplus a_i \quad \text{and} \quad \pi_i : \bigoplus a_i \rightarrow a_i$$

such that

$$\pi_i \circ \iota_i = \text{id}_{a_i} \forall i$$

LEMMA 4.7 (PETER WEYL THEOREM). Let G be a compact group. Then all its unitary representations are completely reducible.

4.1 Representation Theory

IG remove this since we will only be dealing with finite groups? discuss!

Definition 4.8 (Compact Group). A compact group is a group G equipped with a topology such that

$$m : G \times G \rightarrow G, \quad (g, h) \mapsto gh,$$

is continuous.

$$i : G \rightarrow G, \quad g \mapsto g^{-1}$$

is continuous and G is compact as a topological space.

Example 4.9. (1) Every finite group with the discrete topology is compact.

(2)

$$U(n) = \{A \in M_n(\mathbb{C}) : A^* A = I\}$$

is compact for all integer n .

(3) All

$$SU(n) = \{A \in U(n) : \det A = 1\}$$

are compact.

Definition 4.10 (Representation). Let G be a group and V a vector space over a field $\mathbb{F}$ (In our cases $\mathbb{C}$) suffices. A *representation* of G on V is a group homomorphism

$$\rho : G \longrightarrow \text{GL}(V),$$

where $\text{operatorname{GL}}(V)$ denotes the group of invertible linear transformations on V . Equivalently

$$\rho(gh) = \rho(g)\rho(h), \quad \rho(e) = I, \quad \rho(g^{-1}) = \rho(g)^{-1}.$$

The vector space V is called the *representation space*.

Definition 4.11 (Projective Representation). Given a group G , a linear representation of G over $\mathbb{C}$ of dimension n is a map $U : g \rightarrow \mathbb{C}^{n \times n}$.

4.2 Fusion Trees

We consider a fusion tree as the basis that spans $\text{Hom}(R_a \otimes R_b \dots \otimes R_m, \otimes R_{a_j} \dots)$. Since $R_a \otimes R_b$ is semisimple, we have the relationship

$$R_a \otimes R_b \cong \bigoplus_c R_c^{\oplus N_{ab}^c}$$

with embedding (make this formal)

$$g_{c,\mu}^{ab} : R_c \rightarrow R_a \otimes R_b$$

and projections

$$f_{ab}^{c,\mu} : R_a \otimes R_b \rightarrow R_c$$

$$id_{R_a \otimes R_b} = \sum_{c \in L} \sum_{\mu=1}^{N_{ab}^c} g_{c,\mu}^{ab} f_{ab}^{c,\mu}$$

LEMMA 4.12.

$$\text{Hom}(R_{a_1} \otimes \dots \otimes R_{a_N}, R_b)$$

is spanned by ordinary fusion trees.

PROOF. Base case: $N = 1$. Nothing to prove here.

INDuctive case:

Under semisimplicity we get that

$$\text{Hom}(R_{a_1} \otimes \dots \otimes R_{a_n}, R_b) \cong \bigoplus_{c,\mu} \text{Hom}(R_c \otimes R_{a_3} \otimes \dots \otimes R_{a_n}, R_b)$$

□

4.3 Monoidal data

Definition 4.13 ($\text{Rep}(SU(2))$). Let (V, ρ_V) be a representation.

Definition 4.14 (Twists from R moves [Sim23, Equation ?]). Suppose we are in a braided monoidal category with F and R symbols. Then we compute the *twist* (a self braiding) as follows:

$$R_c^{ab} R_c^{ba} = \frac{\theta_c}{\theta_a \theta_b}.$$

(state more possibilities here.)

Example 4.15. Take $sVect$. Our simples are 1 and f . This *braiding* is in fact symmetric. Thus we have that

$$1 = \frac{\theta_1}{\theta_f^2} =$$

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